

Jonah A. Wilkes

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Summary

Aerospace Engineer with a combined B.S./M.E. in Mechanical and Aerospace Engineering from Illinois Institute of Technology (May 2026) and a completed minor in Astrophysics. Engineering focus in dynamics, controls, and estimation with hands-on experience in aerospace systems analysis, multidisciplinary design, mission planning, and research. Skilled in Python and MATLAB for modeling, simulation, signal processing, optimization, and engineering analysis. Project experience spans spacecraft planning and design, GNSS signal processing, and NASA mission concept development. Background includes systems integration, ConOps development, and trade studies through NASA's L'SPACE programs, as well as ongoing contributions to peer-reviewed research in astrophysics and cryospheric remote sensing.

Experience

→ Astrophysics Research Assistant — Department of Physics, IIT

May 2024 – Present | 15 hr/wk

- Analyzed stellar spectra in a Python environment to derive radial velocity curves for 98 binary systems, extract orbital parameters, and evaluate system dynamics.
- Developed Monte Carlo simulations to generate synthetic stellar populations and compare modeled distributions against observed data, performing statistical evaluation of binary-star formation hypotheses.
- Primary author on a forthcoming peer-reviewed publication investigating binarity in Sub-Subgiant stellar systems; work is ongoing post-graduation.

→ NASA L'SPACE Professional Development Training (MCA) — NASA

May 2025 – August 2025 | 15 hr/wk

- Served as the team's primary Mechanical Engineer for a Mars rover mission concept, developing a Siemens NX CAD model that integrated mechanical, payload, thermal, power, and communications subsystems.
- Developed the mission Concept of Operations (ConOps) across four mission phases, coordinating with multidisciplinary team members to support system feasibility and mission execution planning.
- Defined mission and subsystem design under NASA-style project standards, advancing the concept through Preliminary Design Review (PDR).
- Presented the mission concept and design to a Standing Review Board, discussing subsystem integration decisions and overall mission structure.

→ GNSS-R Signal Processing Research Assistant — Armour College of Engineering, IIT

May 2025 – Present | 10 hr/wk

- Developed MATLAB-based processing methods for Global Navigation Satellite System Reflectometry (GNSS-R) data collected over Antarctic glacial surfaces through a NASA Illinois Space Grant-funded research project.
- Processed and evaluated raw reflected satellite signals to extract signal-to-noise metrics and characterize surface reflection behavior using signal processing techniques.
- Performed technical analysis of reflected-signal datasets to support interpretation of glacial surface conditions and ongoing aerospace remote sensing research; work is continuing post-graduation.
- Contributing results to an upcoming peer-reviewed publication.

→ Conceptual Aircraft Design Research Assistant — Armour College of Engineering, IIT

August 2024 – December 2024 | 15 hr/wk

- Developed design and optimization algorithms in Python to accelerate conceptual aircraft design, specifically targeting improvements on the Boeing 737 configuration.
- Built tools using Latin Hypercube sampling to generate and evaluate large parametric design spaces, enabling rapid assessment of performance trends across multiple design variables.
- Consolidated the analysis and optimization workflow into an interactive application that streamlined trade studies, improved comparison of candidate designs, and reduced design cycle time.
- Awarded First Place at the Fall 2024 Armour R&D Expo for this research project.

→ NASA L'SPACE Professional Development Training (NPWEE) — NASA

May 2024 – August 2024 | 10 hr/wk

- Contributed to a complete NASA-style mission proposal within a multidisciplinary team, working under NASA proposal standards and contributing mission design, technical analysis, and written proposal content.
- Developed substantial technical material for a proposal reviewed through official NASA channels, helping define mission concept content and written documentation.
- Served on a NASA-style proposal review board, evaluating peer submissions and providing technical feedback on mission design approaches.

Education

Fall 2021 – May 2026

Illinois Institute of Technology

Bachelor of Science in Aerospace Engineering

Master of Science in Mechanical and Aerospace Engineering

Minor in Astrophysics

Chicago, IL

ABET Accredited

ABET Accredited

Completed Fall 2024

Undergraduate GPA: 3.851/4.0 **Graduate GPA:** 3.709/4.0

Graduated: May 2026

Armour College of Engineering Dean's List: Every semester enrolled (Fall 2021 – Spring 2026)

IIT STEM Scholarship Recipient

Relevant Courses: Spacecraft Dynamics & Design, Aircraft Design, Aerospace Propulsion, Navigation Systems, Optimal State Estimation, Optimal Control, Systems Analysis and Control, Advanced Dynamics, Aerodynamics, Aerostructures, Compressible Flow, Computational Mechanics.

Technical Skills

Programming & Analysis:

Python, MATLAB, Linux, Monte Carlo simulation, statistical analysis, data visualization, signal processing, optimization, technical modeling

Aerospace Modeling & Simulation:

STK, Simulink, AVL, XFLR5, OpenRocket, spacecraft dynamics modeling, orbital and mission analysis, Lambert solvers, porkchop plot analysis

Systems Engineering & Mission Design:

Systems integration, ConOps, trade studies, mission planning, design reviews, requirements development, sensitivity analysis, multidisciplinary design

CAD & Design Tools:

Siemens NX (Certified), SolidWorks, 3D assemblies, mechanical drawings

Controls, Estimation & Navigation:

Optimal state estimation, optimal control, navigation systems analysis, dynamics and control modeling

Extracurriculars

Illinois Tech Baseball

Fall 2021 – Spring 2026

5-year varsity athlete and starting catcher; developed leadership, teamwork, and communication in a competitive DIII team environment

Illinois Tech Rocketry Club

Fall 2022 – Spring 2026

Member; collaborated on engineering projects and rocketry design efforts